

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on a simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies

these implications; one model satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a pre-registered decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent

15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which; each result
23 names a measurement that phase data alone cannot supply.

24 The objection that synchrony does not fix content is older than this work.
25 Critiques of the temporal binding hypothesis argued that synchrony, where
26 it occurs, cannot say which features belong together or what they jointly
27 represent: that grouping has to be computed before synchrony could sig-
28 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
29 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
30 and a pigeonhole count. What formalization adds to the informal argument
31 is quantitative and bookkeeping content. It gives the rate at which a co-
32 herence bound on content weakens with population size, the population at
33 which it stops constraining anything and how far chaining through overlap-
34 ping patches extends it, the diameter within which one contracting map can
35 reconstruct, and the number of codes a separated family of states requires.
36 It also gives a complete inventory of the premises a synchrony-to-content
37 argument consumes, each of which names a measurement, together with one

38 model in which all of them hold at once.

39 The familiar noisy mean-field oscillator supplies a stationary coherence
40 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991); we add a minimum
41 cost for changing coherence quickly. A Lean 4 development checks the impli-
42 cations, and a joint witness satisfies the eight connecting hypotheses of the
43 formal composition in a specified model. It proves neither that cortex satisfies
44 those hypotheses nor that the glued state is experienced; that identification
45 needs its own endpoint.

46 The empirical question is which measurements could support or discon-
47 firm the proposed connections. A phase resultant can miss a locally ordered
48 winding field; stationary phase distributions can leave the coupling mecha-
49 nism unidentified; and coherent adaptation can lose alignment with a desig-
50 nated content template. We therefore distinguish stationary branches from
51 finite observed onset, and compare prespecified field and connectivity onset
52 models on held-out emergence recordings. The comparison tests a specified
53 spatial prediction, not the presence of experience. Section 10.2 states the
54 comparison that bears on conscious content directly.

55 Sections 2–5 develop the three requirements; Section 6 applies their mea-
56 surement obligations to a digital candidate. Section 7 states what the formal
57 composition establishes, and Sections 8–9 examine observables and the cor-
58 tical test. Table 1 lists the connecting hypotheses the article argues from;
59 Supplemental Material gives the full chain, derivations and controls. For a
60 theoretical article the Methods are the definitions, proofs and simulations.

Each result is stated in the section that uses it, Supplemental Material Table S1 maps it to its Lean declaration, and every simulation’s model, parameters and controls are given in Supplemental Material. The Results are the theorems of Sections 3–5 and the numerical controls of Section 8.

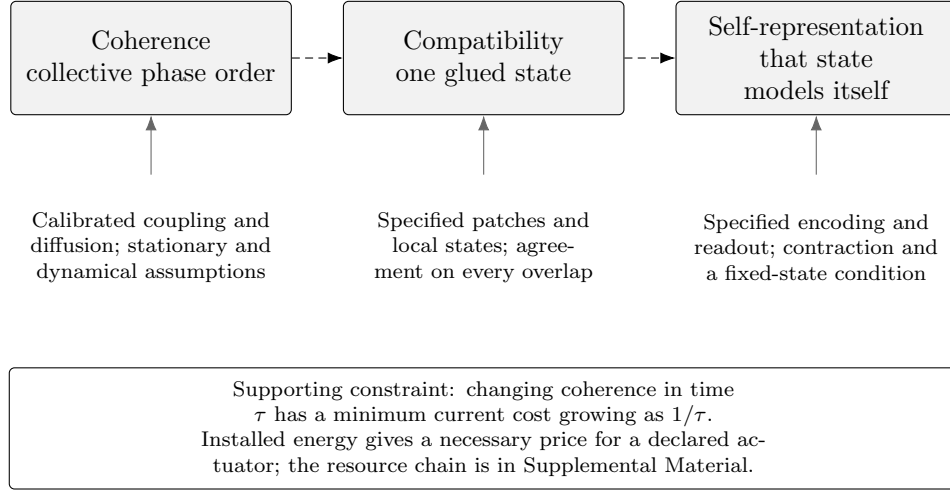


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

64

2 Limits of phase order

Several limits hold without the cortical hypothesis. Equal phases can coexist with incompatible local descriptions; phase order constrains shared content only through encoders related on overlaps. A winding field can have zero

69 global resultant while remaining locally ordered. A constant readout can have
70 a unique fixed point without reconstructing its input. A region encodes only
71 the states it distinguishes, and by a declared deadline a site can report only
72 changes inside its causal past. These limits reflect familiar phase oscillator
73 and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008;
74 Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one
75 beside the condition or measurement it motivates.

76 **3 Coherence: physical coupling and its cost**

77 **3.1 Dynamics and the stationary threshold**

78 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

79 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
80 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
81 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
82 quency heterogeneity and diffusion are distinct parameters. The stationary
83 calculation below uses identical frequencies and this scalar coupling. Ap-
84 plying it to a spatial kernel requires an aggregation rule and a check of the
85 resulting approximation.

86 For identical frequencies in their rotating frame, the positive normalized

87 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

88 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 89 mal derivation uses the absence of stationary current for a periodic gradient
 90 drift to obtain this density, then verifies normalization and self-consistency.
 91 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 92 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 93 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

94 Figure 2 shows this stationary branch.

95 Linearization about the uniform density gives rate $K/2 - D$ for the first
 96 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 97 exactly above threshold. About the coherent density, the quadratic form
 98 is strictly dissipative transverse to collective rotation, whose rate is zero.
 99 These linear estimates concern the limiting equation. Relating them to finite
 100 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 101 tion; exchange symmetry alone does not establish the needed independence
 102 (Sznitman, 1991).

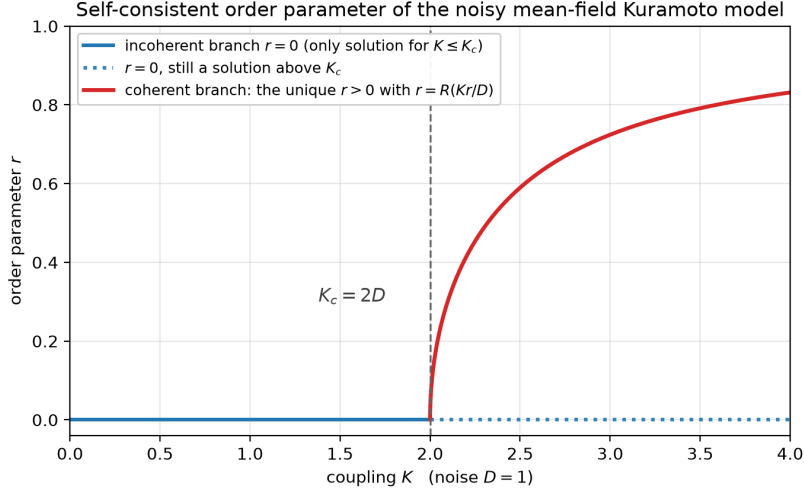


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

103 3.2 A speed limit on changes of coherence

104 One thermodynamic statement bears on coherence directly. For a smooth
 105 positive periodic phase density with probability current J and diffusion D ,
 106 the current functional $\sigma_J = \int J^2 / (D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

107 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
 108 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
 109 has no positive floor. Read the other way, a current budget C sets a minimum

110 duration $\tau \geq [\Delta \arcsin r]^2 / (DC)$ for a given change. This bears on ignition-
 111 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
 112 2014), and on recovery latency after anaesthesia: each has a floor set by the
 113 current available to drive it. The bound is a heat or power statement only
 114 when J is identified with a physical flux, a reservoir at a stated temperature
 115 is specified and local detailed balance holds for the steps concerned; only
 116 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
 117 Shenker, 2000). Installed energy supplies a separate, necessary price: for
 118 a declared actuator whose response modes carry installation prices, mean
 119 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
 120 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
 121 is known. Supplemental Material gives both derivations, the register and
 122 feedback budgets, and the sensitivity of the metabolic comparison to its
 123 assumptions.

124 **4 Compatibility: from coordinated activity** 125 **to a global description**

126 **4.1 Why common timing is insufficient**

127 The formal setting is a sheaf: each patch of the substrate carries local de-
 128 scriptions, and restriction maps cut a description on a patch down to any
 129 smaller patch. A *section* is a description on one patch, and a *global section* is

130 a description on the whole substrate. For overlapping patches U and V , com-
131 patibility means that restricting either local description to $U \cap V$ gives the
132 same object. Descriptions can agree on shared quantities while contributing
133 different information elsewhere. Compatible sections of the specified sheaf
134 have a unique global extension. On a finite cover, local finite spatial mea-
135 sures agreeing as measures on every overlap also have a unique finite-measure
136 extension whose sheaf image is the glued section. No mass-one normalization
137 is imposed.

138 In the cup example, visual and reaching descriptions must express a
139 shared quantity, such as cup location, in common coordinates. Restriction
140 retains that quantity while dropping details particular to each description.
141 Agreement permits surface appearance and hand configuration to contribute
142 to one situation. The mathematical counterexample assigns the same phase
143 to every patch while two local mass profiles disagree on a shared site. Those
144 profiles have no common extension. Because they are chosen independently
145 of phase, synchrony places no constraint on their disagreement.

146 This distinction limits the biological interpretation. Receptive territories
147 and content domains must be related to spatial patches, and the descriptions
148 must support the required extension. Matching overlap marginals of de-
149 coded probability laws need not determine a unique global joint distribution
150 (Abramsky & Brandenburger, 2011).

151 The cover is itself a choice, and the construction constrains it in one way
152 only: synchronized patches agree where they overlap. Where they do not

153 overlap there is nothing to compare, so a cover by pairwise disjoint patches
154 satisfies that obligation for *every* assignment of local descriptions and phases
155 and the gluing theorem still returns a unique global section, which is the
156 tuple it was handed. Membership of the class then reports nothing about the
157 system beyond the patches covering it. What a gluing claim asks of a cover
158 is therefore a non-degeneracy the theorem does not assume: that distinct
159 patches meet, and that what they share is a quantity fixed independently of
160 the agreement it is being used to establish.

161 The cup example exposes a second gap, between two kinds of overlap.
162 The sheaf is built over a spatial substrate, so its patches overlap where they
163 share sites. Visual and motor territories that agree on cup location overlap
164 in what they describe, not in where they lie, and the construction supplies
165 no relation between the two. One route recasts the construction over content
166 variables: each region becomes a partial decoder whose domain is the set of
167 variables it estimates, and two regions overlap exactly where they estimate
168 a common variable. Gluing and approximate selection hold on such a cover,
169 and there compatibility is a consequence: regions decoding a variable within a
170 tolerance agree on it within twice that tolerance, so only agreement beyond
171 fidelity is evidence. The phase-to-content bounds concern spatial patches
172 and do not transfer. One principle selects such a cover without presupposing
173 the agreement it is used to test. Assign each region the variables that can
174 be decoded from its activity above a declared criterion, on data held out
175 from the agreement test, and let two regions overlap on exactly the variables

176 decodable from both. Decodability is fixed by what each region carries before
 177 any comparison is made, so the cover cannot be chosen to make agreement
 178 come out. It is non-degenerate exactly when its nerve is connected, meaning
 179 that every region is linked to every other through a chain of shared decodable
 180 variables, and recordings can fail that condition. The principle has three
 181 inputs, the candidate variables, the criterion and the decoder class, and each
 182 has a rule that does not consult agreement.

183 The candidate variables are those the experiment manipulates and the
 184 task makes behaviourally relevant, fixed before recording. Data-driven alter-
 185 natives, such as directions of maximal shared variance between two regions,
 186 select for the agreement under test and are excluded. The criterion becomes a
 187 statistical one: a region encodes a variable when its cross-validated decoding
 188 exceeds a permutation null at a pre-registered level, and results are reported
 189 across a declared range of levels. The level need not be exact, because two
 190 accurate decoders agree automatically (Eq. (13)); the statistic of Section 10.2
 191 is agreement in excess of what each region’s accuracy implies, which remains
 192 valid at any admitted level. The decoder class is bounded on both sides.
 193 Unrestricted nonlinear decoders recover a variable from any region carrying
 194 information about it, so nearly every pair of regions overlaps and the cover
 195 again constrains nothing. The positive bound of Section 4.2 requires encoders
 196 with a declared Lipschitz constant L , which enters δ_r directly, so the class
 197 should consist of L -Lipschitz maps; linear readout, the usual operationaliza-
 198 tion of information explicitly available to downstream neurons (DiCarlo &

199 Cox, 2007; Kriegeskorte & Diedrichsen, 2019), is one such class. Within the
 200 admitted family, the decoder and the distance in which content discrepancy is
 201 measured are chosen by held-out decoding accuracy and never by agreement.
 202 Neither is neutral: which similarity measure best describes neural activity
 203 differs across tasks (Bobadilla-Suarez et al., 2020). What remains is that the
 204 cover is the experimenter’s. Whether cortex realizes unity over the chosen
 205 variables is the empirical claim under test, and no choice of decoder settles
 206 it. The correlate is therefore operational relative to these pre-registered in-
 207 puts, and carrying the formal results to such a cover is the recasting sketched
 208 above.

209 **4.2 When coherence does constrain content**

210 A declared relation between phase and content gives a positive result. Let
 211 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 212 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 213 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 214 meaning that an encoder’s output moves by at most L times the chord
 215 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 216 j overlap. Encoders may vary across sites; their equality on overlaps is a
 217 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 218 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors’ squared deviations from their
 219 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 220 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap

221 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (5)$$

222 Here discrepancy is the norm of the difference, and r and N are read on
223 whichever population the bound is applied to. That choice is what decides
224 whether the estimate says anything. A patch carries an order parameter
225 of its own, and reading the bound there carries that patch's resultant and
226 that patch's site count, so a large population observed through small patches
227 is constrained by the small count. Read instead on the whole array, the
228 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
229 from an average over N sites. The order cannot be improved in general:
230 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
231 observations agree exactly at either scale. A shared scalar encoder is a special
232 case. Dropping the overlap relation permits disagreement even at perfect
233 locking. A common nonexpansive linear projection preserves agreement, but
234 agreement of a projection does not establish agreement of the full vector.

235 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
236 constrains anything only where the right-hand side falls below that. At the
237 sheet's own patch order, the bound is informative for at most 10 sites. Three
238 conditional results decline the step that puts $\sqrt{N}$ there. A pair count bounds
239 the fraction of disagreeing pairs at any population size but names no pair.
240 A chain across the sites that overlapping patches share compares distant

territories through the patches between them, with error growing linearly in hops rather than with population: on the same sheet it stays informative for 19 hops, 0.312 mm. A spectral gap removes the population count at a frequency-locked configuration, given a declared algebraic connectivity of the coupling graph. Territories whose agreement unity concerns lie centimetres apart, beyond the chain's reach, and no measured connectivity yet supplies the spectral route, so across them the coherence-to-content link is the bridge assumption E78 (Table 1). Supplemental Material gives the three derivations and the sheet's numbers.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i + \eta o_i$. Content and observation discrepancies bounded by ε and δ give a next discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, reduce an initial mismatch. Approximate observations leave a nondecaying term in the error bound. Taking a uniform bound on Eq. (5) connects this update to the phase process.

4.3 What approximate agreement determines

Exact gluing requires exact restrictions. For nonnegative profiles on a finite substrate, suppose every overlap discrepancy is at most ε in uniform distance.

261 Weights that vanish outside their patches and sum to one at each site select
 262 a global profile by weighted averaging. The selection lies within ε of each
 263 patch there, and any two such selections differ by at most ε , independently
 264 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 265 $L\varepsilon$ across selections.

266 At zero tolerance, the selection is the unique exact extension. At every
 267 positive tolerance there are families with no exact extension; selection then
 268 gives an approximation whose dependence on weights is bounded. Supple-
 269 mental Material gives a three-site witness attaining the bound. The mass-
 270 profile construction uses the scalar case of the content dynamics and addi-
 271 tionally requires nonnegativity of the initial profiles and observations. The
 272 general vector result supplies no identification with the spatial-measure sheaf.

273 These constructions concern declared objects: a specified cover, contents
 274 valued in a normed space, and encoders related on overlaps. They select
 275 neither the neural cover nor the decoder, and identifying their contents with
 276 perceptual content, or establishing that the identification bears on experi-
 277 ence, is a further modelling task; Section 8 sets out what a measurement of
 278 it would have to establish.

279 5 Self-representation: a state that models it- 280 self

281 5.1 The reflexive construction

282 Compatibility gives a global state. Reflexivity relates it to an encoding within
283 that state. Equip global sections with a complete metric. *Restriction reso-*
284 *nance* is the additional condition that the encoding is the state’s own restric-
285 tion to a region $\mathcal{A}$, so that the region encodes the state by being part of it.
286 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

287 The region and readout are fixed across the state space. If F contracts and
288 fixes the section s_* formed from the local descriptions, Banach’s theorem
289 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
290 (Table 1 gives the full composition and its edges) supplies the contraction
291 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
292 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
293 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
294 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density
295 grows (Section 3), a rate about a different state from the one F acts on,
296 and no result derives a Lipschitz law for an encoding or readout from the
297 phase dynamics. Constructing a map with that factor remains a modelling

obligation. A fixed contraction can reconstruct accurately only a restricted family of states: if its error is at most ε throughout that family, any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

In the cup example, the global state includes the represented scene and the perceiver's relation to it. A local encoding must retain enough information for the readout to recover that modelled state; cup position alone would be insufficient if hand configuration and viewpoint could not be reconstructed.

For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.

316 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 317 existing self-prediction map.

318 This is an additional accuracy criterion on a family, with no requirement
 319 for multiple exact fixed points. For the same fixed contracting map used by
 320 E89, however, the diameter bound above is an additional restriction: at zero
 321 error the family contains at most one metric-distinct state. A broad family
 322 needs an input-conditioned map, one contraction F_u per input u . If F is
 323 L -Lipschitz in u , members with inputs u, v lie within $(2\varepsilon + L d(u, v))/(1 - \Lambda)$
 324 and the fixed points within $L d(u, v)/(1 - \Lambda)$, so a self-state can follow the
 325 scene. The code criterion then binds within each input: a readout storing
 326 one state per input reconstructs without encoding anything. No physical
 327 F_u is constructed, so under E89 the minimal-self correlate is a condition on
 328 one self-state. A blind encoder or constant readout fails the criterion for
 329 separated states despite having a unique fixed point.

330 Where the encoding is read as the state's own restriction to a region, the
 331 criterion becomes a condition on the region. Restriction resonance requires
 332 the encoder and that restriction to identify the same pairs of states, so a
 333 boundary reconstructing two relevant states more than 2ε apart while its
 334 region reads them identically is not resonant. Read forwards, a resonant
 335 boundary that reconstructs a pairwise separated family restricts injectively
 336 on it. Which states a region reads identically is fixed by where the region
 337 sits, before an encoding is chosen, so the resolution the family demands is
 338 the region's and no readout supplies it. The implication runs one way: a

339 region that does resolve the family is not thereby shown to carry a resonant
 340 encoding, and E89 supplies no such resolution.

341 Resonance and resolution are jointly satisfiable without a local fold. Take
 342 the region to be the whole substrate and use the identity readout: restric-
 343 tion is the identity, so this boundary is resonant and reconstructs every de-
 344 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 345 E89’s strict contraction premise on a nontrivial metric space. A properness
 346 condition on the region is therefore a separate requirement, and the fixed-
 347 contraction diameter bound remains in force when E89 is also required. With
 348 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 349 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 350 so the family is no larger than the δ -packing number of what the region car-
 351 ries. At the whole substrate it says nothing, because the region’s range is
 352 everything. The gain L is a declared input, derived here for no physical
 353 readout, and the bound weakens without limit as the gain grows.

354 **5.3 Agreement after a change takes communication**

355 Snapshot compatibility does not specify how quickly descriptions respond
 356 when a shared quantity changes. In a synchronous network whose updates
 357 read only the site’s own state and declared neighbours’ messages, two initial
 358 configurations agreeing on a site’s causal past through T rounds produce the
 359 same state there at round T . The deadline consequence is the one Section 2
 360 states, and it follows from the reconstruction bound in Section 5.2: a report

361 from that site is a readout of a code that the two interventions leave identical.

362 A three-site delay line witnesses failure after one round and exact report-
363 ing after two; cutting the path prevents delivery at every round. A report
364 may still happen to be correct about one fixed world. The bound concerns
365 responsiveness across interventions.

366 Throughout this section *prediction* names a reconstruction map, and the
367 relevant family of states, the metric, the tolerance, the communication graph
368 and the deadline are all declared inputs, as are the represented variables and
369 the fidelity counted as accurate. Reading the map as temporal neural pre-
370 diction would require dynamics; reading the resulting code count as physical
371 memory capacity would require a hardware model, which a state-dependent
372 readout would have to extend.

373 6 What the conditions ask of a digital candi- 374 date

375 A claim that a digital system, such as a large language model (LLM) run-
376 ning on graphics processing units (GPUs), has unified content or represents
377 itself commits its maker to the measurements this account asks of cortex.
378 The maker must identify the shared quantities the system’s parts describe,
379 show that decoders fitted independently on those parts agree about them,
380 and show that a declared local code separates the relevant family of states
381 and survives intervention. The indicator-properties approach assesses arti-

382 ficial systems against computational properties drawn from theories of con-
 383 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
 384 a measurement rather than an architectural property. A digital system is
 385 also where they can be measured today, with full access to its state and
 386 exact interventions on it: fitting decoders for one shared quantity on two
 387 separate layers, heads or token positions, and testing their agreement and
 388 whether reconstruction survives intervention, would carry out the proposed
 389 measurement directly.

390 One constraint follows from the architecture alone. Within one forward
 391 pass of a causally masked model, position t reads only its causal past, so an
 392 intervention outside that past cannot be reconstructed at t by that step. This
 393 is the deadline result of Section 5 applied to the mask, and it bounds which
 394 variables a position can encode about the rest of the sequence within one
 395 step. A second constraint needs a premise the architecture does not supply:
 396 a linear encoder and readout factoring through d coordinates pay reconstruc-
 397 tion error at least the discarded spectral tail of the coupling operator, but
 398 query/key dimension does not supply such a bottleneck for softmax atten-
 399 tion. These results rank no architecture family, imply no general inferiority of
 400 GPU hardware and settle no question of machine experience. Supplemental
 401 Material gives the bottleneck derivation, a synthetic two-view task with its
 402 controls, the information ceiling of token outputs and the hardware scope.

403 7 The formal composition

404 The Lean development composes the three requirements into one conditional
405 theorem. Its passive and feedback branches each accept eight connecting hy-
406 potheses, running from finite capacity through a register, dissipation and a
407 continuum coupling to the field, coherence, the cover and the self-prediction
408 fixed point. Each edge is supplied rather than derived from the one above it,
409 and the inferential content of the composition is its last step, where contrac-
410 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
411 article argues from; Supplemental Material gives the full chain, its thermo-
412 dynamic edges and the register ledger of the feedback branch. Both branches
413 also take a cover carrying exact overlap compatibility and a reflexive bound-
414 ary specifying encoding and readout, and conclude that the glued section is
415 the unique fixed point of self-prediction. Composing the edges establishes
416 that the premises are consistent, named in one place and machine-checked
417 together; it does not establish that self-representation follows from coher-
418 ence. Joint witnesses satisfy every edge of each branch in a specified model,
419 and establish no common cortical mechanism. The development declares no
420 additional axioms beyond its libraries' foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover’s nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve’s diameter; where the configuration is frequency-locked it carries through the coupling graph’s algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

421 8 What the observations establish

422 The composition holds under its premises. Whether observation can estab-
423 lish those premises is a separate question, and the three requirements it
424 connects concern different objects: phase dynamics, local descriptions and
425 internal encodings. Their observables must retain these distinctions. The
426 following controls show which mechanisms remain indistinguishable, which
427 temporal approximations fail, which states one observable cannot separate,
428 and what independent content measurements must establish. Each is a finite
429 numerical regime on a declared model, with full parameters and controls in
430 Supplemental Material, and none of them calibrates a cortical quantity.

431 8.1 Stationary distributions leave the mechanism un- 432 determined

433 Equation (2) gives a concentration–coherence relation with no fitted shape
434 parameter. Estimate r from circular resultant length and a from density
435 shape: inverting R from the measured r would make agreement automatic.
436 Even independently estimated agreement identifies a statistical family shared
437 by different mechanisms.

438 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
439 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
440 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
441 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

442 and diffusion D_0 have that same density. Their stationary distributions co-
 443 incide window by window despite different causes.

444 Resolved dynamics supply additional information. In the ideal model
 445 with continuous latent paths, quadratic variation accumulates at rate $2D$
 446 and conditional increment means constrain drift; a filtered observation of
 447 the phase needs its own observation model before either is read off. Separat-
 448 ing endogenous coupling from a matched common drive additionally requires
 449 a measured drive or controlled perturbation; identifying electromagnetic cou-
 450 pling requires evidence specific to that interaction. Supplemental Material
 451 gives the equivalence construction and its numerical check. Phase statistics
 452 pooled over time and over bipolar pairs cannot supply these estimates: on
 453 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 454 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 455 that removes the common rotation and a montage whose effect on spatial
 456 phase is modelled; Supplemental Material reports this estimator exercise and
 457 the dependence-aware design.

458 **8.2 Finite ramps change the observable onset**

459 The square-root law in Eq. (3) describes stationary solutions, and a chang-
 460 ing coupling needs time for the phase distribution to approach them. In a
 461 deterministic mean-field control crossing the threshold, none of the scanned
 462 rates meets the declared tracking tolerance at onset, so a slow control vari-
 463 able supplies no general guarantee of branch tracking. In a finite-population

464 ramp study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps de-
 465 lay escape. The fitted delay exponent 0.444 has a replica-bootstrap interval
 466 $[0.398, 0.496]$ that excludes the ideal $1/2$, a deterministic reference shows that
 467 the escape criterion changes the fitted exponent, and no common asymp-
 468 totic exponent is established. Delayed passage through a noisy bifurcation
 469 (Berglund & Gentz, 2002) makes the apparent exponent depend on ramp
 470 speed and fitting window. Supplemental Material gives the sweep and its
 471 figure, the fluctuating-coupling and population-size controls, and the signed-
 472 network crossing bracket.

473 **8.3 A winding state leaves the order parameter am-** 474 **biguous**

475 The global order parameter r discards spatial arrangement. On a periodic
 476 sheet, a phase field can wind once around the sheet and have $r = 0$ while
 477 neighbouring sites remain locked. At identical frequencies, every such wind-
 478 ing is stationary for an isotropic symmetric kernel: contributions at opposite
 479 separations cancel. It violates the quarter-turn initial-spread premise of the
 480 convergence result, which is why that result can still select the uniform state.

481 In the spatial control, the winding persists below the interpolated de-
 482 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 483 and winding starts under the same coupling, diffusion and kernel end at
 484 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-

485 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 486 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 487 realizations. These are finite controls, not a cortical wave identification or a
 488 derived transition.

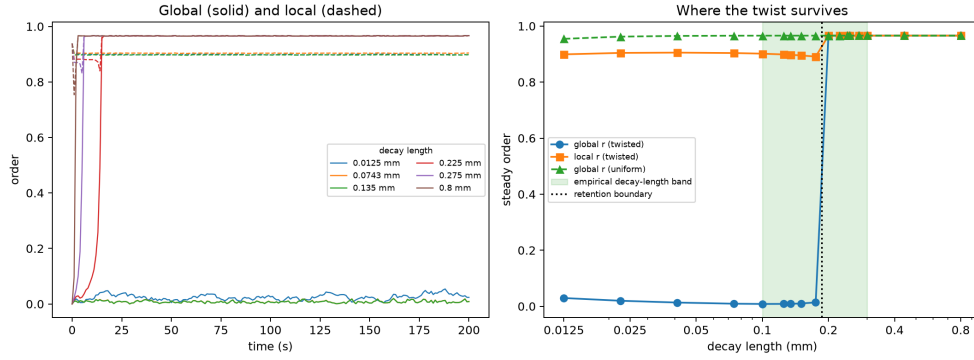


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

489 Linearization about a winding gives explicit mode rates, which concern
 490 linear stability rather than the measured retention boundary. An amplitude
 491 extension adds an existence condition under which a broad kernel can remove
 492 a winding that the phase-only equations retain as a stationary state.

493 Neither model turns these states into measured cortical spirals. The pe-
 494 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has

no propagation speed, and the amplitude variable has no calibrated relation to a recorded envelope. Supplemental Material gives the spectral, amplitude and defect results. Locally, Eq. (5) can still bound content disagreement on a winding patch when shared encoders are supplied, although its global $r = 0$ form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters. Supplemental Material proves descent of this objective for a specified gradient flow at fixed phases and distinguishes it from entropy production.

The finite control co-evolves noisy phases and symmetric, nonnegative

couplings at fixed total resource. Its template is a diagnostic covariance of latent signals, never a training input. Across three illustrative seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and in-

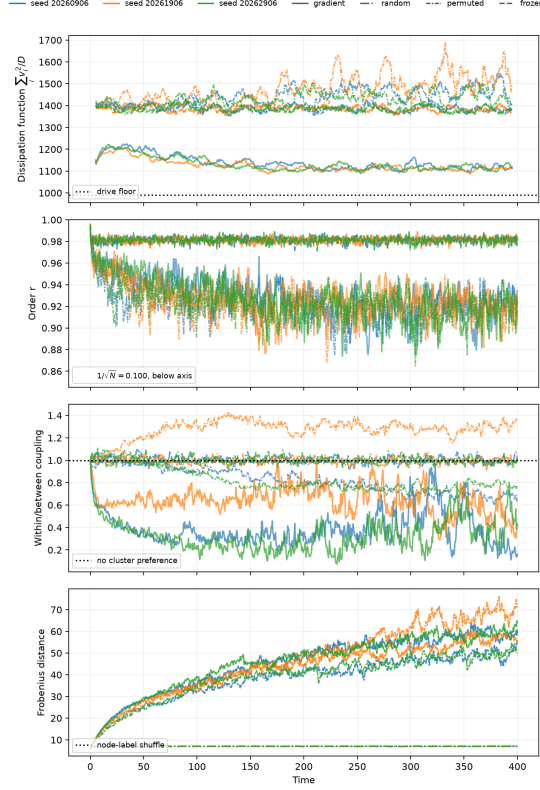


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

compatible examples. An implementation on synthetic data separates these cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. Section 4 gives the decodability rule that fixes the cover without consulting agreement.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what

559 observations must distinguish before a phase trace can support this identifi-
560 cation.

561 **9.1 Physical identification and calibration**

562 The cortical hypothesis identifies part of effective coupling with endogenous
563 extracellular electromagnetic interactions. Fields can modulate spike timing
564 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
565 propose coordination of distributed neural information (Pinotsis & Miller,
566 2022; McFadden, 2020). Phase organization under propofol motivates study-
567 ing this regime (Bastos et al., 2021), but does not measure coupling.

568 The formal field predicate requires a jointly continuous kernel, a normal-
569 ized substrate, coupling equal to the kernel average, and no single site pro-
570 viding more than half that strength. Toy constructions satisfy it. A cortical
571 realization must specify the scale at which the kernel and its reduction ap-
572 ply: calcium imaging reports different state-dependent modularity patterns
573 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

574 Cortical columns are a serious candidate for the oscillator index. They are
575 delineated by response properties and anatomy rather than by the agreement
576 they would be used to establish (Mountcastle, 1997), and a phase can be
577 estimated from a column-sized territory at the electrode spacings used for
578 cortical wave measurements (Townsend et al., 2015). They are a poor content
579 cover: non-overlapping columns form a disjoint cover (Section 4), much of
580 association cortex has no accepted columnar parcellation, and the column’s

standing as a canonical unit is contested (Horton & Adams, 2005; Rakic,
 2008). The content cover is the decodability cover of Section 4, and the
 bridge assumption E78 records its identification with the oscillator index.

The spatial reduction is also testable. A row-sum aggregation at zero
 diagonal tracks the uniform-coupling control over the tested spatial decay
 range, while frequency heterogeneity displaces the operational threshold. The
 fit window biases both estimates relative to $2D$; agreement with the control
 is therefore the supported conclusion. Supplemental Material reports the
 reduction, its numerical error and the distinct assumptions needed to identify
 the kernel average with the scalar coupling of the installed-energy condition.

The field commitment requires physical evidence beyond a choice of math-
 ematical representation. The support comparisons in Supplemental Material
 concern specified support constraints and supply neither that evidence nor
 energy prices.

A spike-time shift alone supplies no coupling rate. The dimensionless
 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

This rate describes how induced timing shifts accumulate into drift. Nei-
 ther it nor the installation factor κ follows from a heat bound. Testing the
 physical threshold requires independent estimates of the interaction kernel,

its aggregation and diffusion in the same cortical regime.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (10) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives a deterministic tracking approximation that fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

This is a deterministic tracking approximation from Eq. (3) at fixed D ; the crossing controls in Section 8.2 show that it fails as an observed-onset ap-

proximation under the tested initialization. A separation of geometry and phase time scales can support tracking away from threshold. The awakening hypothesis needs a calibrated dynamical prediction for an operational onset above the finite-size fluctuation floor. Crossing K_c alone does not supply a recovery latency.

9.3 A protocol for the awakening experiment

Neural inertia describes a behavioural barrier and hysteresis between anaesthetic induction and emergence (Friedman et al., 2010); recovery can pass through metastable activity states (Hudson et al., 2014; Proekt & Hudson, 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field mechanism from a synaptic one. A possible discriminator is spatial onset relative to independently estimated local field coupling and network connectivity. The present mean-field model has no spatially varying κ and derives no nucleation order, so a field-based onset map needs a separate model specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions i and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements, and a connectivity score c_i from a separately estimated structural network. For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

642 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

643 Predicted onset is the first crossing of a region-specific threshold fixed from
644 pre-onset fluctuations. Fit the two scales on training sessions, then com-
645 pare held-out onset times and site order under the same observation model.
646 Larger field scores predict earlier onset only when baseline order and diffu-
647 sion do not reverse the ranking. This declared approximation omits interre-
648 gional coupling and saturation; a successful ranking would support its field
649 covariate, not prove ephaptic causation. Supplemental Material specifies the
650 controls and failure conditions.

651 A within-subject protocol can manipulate emergence rate, with spatially
652 resolved recordings to compare the proposed field and connectivity models:

- 653 1. **Manipulate emergence rate.** Vary the propofol infusion taper
654 across sessions within the same subject, recording drug concentrations
655 or a validated pharmacokinetic proxy to quantify the manipulation.
- 656 2. **Measure behavioural recovery.** Record time-to-response to a stan-
657 dard stimulus as the independent behavioural endpoint, rather than
658 estimating ΔK directly.
- 659 3. **Resolve spatial onset.** Identify anatomical locations where coher-
660 ence first crosses the baseline-scaled threshold and score the two maps

661 of Eq. (12) on held-out sessions. The two covariate maps are fixed
662 before the outcome sessions are inspected.

663 **4. Match physical controls.** Confirm that the observed coherence-to-
664 amplitude relation holds under the specific local diffusion D and ob-
665 servation noise. The model’s quadratic-variation rate $2D$ applies to
666 the latent phase; a band-limited analytic phase is smooth below its
667 correlation time, and on a constructed field its squared-increment rate
668 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
669 fore requires an explicit observation model for filtering and sampling.

670 Predeclare the recording window, regional fluctuation thresholds, and a
671 shared timing-error model. Fit the two scales and that observation model
672 on training sessions only; on held-out sessions, score the observed site-onset
673 times and sites that never cross by the same regional predictive rule. Report
674 site order as a secondary outcome and behavioural time-to-response sepa-
675 rately. Sessions without a crossing are censored observations, not zero delays
676 or exclusions. Compare the two scores only where independently estimated
677 field and connectivity maps predict distinguishable onset patterns at the
678 recording resolution. A held-out field-model advantage would support that
679 specified spatial predictor; it would not identify an ephaptic cause, shared
680 content or experience. Supplemental Material gives the controls and pilot
681 quantities needed to judge whether the comparison is informative.

682 10 Discussion

683 10.1 Unity and a minimal self as interpretive motiva- 684 tion

685 We propose the state glued from compatible local descriptions as a formal
686 correlate of conscious unity, and its accurate internal reconstruction as a cor-
687 relate of a minimal self. In the cup example, this would mean one description
688 that preserves the cup’s relation to the perceiver as well as the scene. Com-
689 patibility permits distinct contents and uncertainty; the shared quantities
690 must be chosen from independent evidence about what each region encodes.
691 The reconstruction need not be a verbal or autobiographical account.

692 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
693 ing the subject of an experience, present before any narrative or reflective
694 self-description (Gallagher, 2000). Two of its features motivate the proposed
695 correlate. It is not a separate object of experience but a perspective built
696 into the experienced scene; correspondingly, the reconstruction is not a sec-
697 ond representation beside the glued state but a map whose fixed point is that
698 state, read through a part of it. It is immediate rather than inferred from
699 a higher-order state; correspondingly, the fixed point needs no further state
700 that represents the first. The correspondence is structural and does not show
701 that a fixed point is experienced as a perspective. Self-model theory locates
702 the phenomenal self in a transparent model a system has of itself (Met-
703 zinger, 2003), and interoceptive-inference accounts ground it in prediction of

704 the body’s own states (Seth, 2013). The present condition is narrower than
 705 either: it asks only that a declared local encoding recover the global state,
 706 including perceiver-relative variables, at a declared fidelity, and it says noth-
 707 ing about transparency or about which bodily variables must be modelled.
 708 Either account could supply the family of states and the fidelity that the con-
 709 dition leaves as inputs. Under E89 that family is one state at zero error and
 710 a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2), so under
 711 E89 the condition constrains a single self-state; a self that persists through
 712 change needs an input-conditioned map, for which no physical readout is
 713 given. Its proposed relation to experienced perspective remains an inter-
 714 pretation. Neither identification is among the connecting hypotheses of the
 715 composition (Table 1), and satisfying them would not measure experience.
 716 The unity proposal is also only as definite as its cover. The gluing theo-
 717 rem holds for whatever cover it is given, a disjoint cover satisfies it trivially,
 718 and the cover of interest overlaps in content rather than in space. Section 4
 719 proposes fixing that cover by decodability on held-out data, which excludes
 720 covers chosen for agreement and makes non-degeneracy a testable property
 721 of the recordings. Its variables, criterion and decoder class are pre-registered
 722 by rules that never consult agreement.

723 10.2 A comparison that could disconfirm the link to 724 content

725 The framework predicts that conscious content tracks compatibility, not co-
726 herence. The comparison must also separate compatibility from how well
727 each region encodes the stimulus, because two accurate decoders agree auto-
728 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
729 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

730 Consciously perceived stimuli are usually decoded better, so higher agreement
731 on those trials would follow from single-region fidelity alone. At matched
732 single-region error, agreement differs only through the covariance of the two
733 errors, which is the term compatibility bears on: two regions describing one
734 content should err together.

735 Present matched stimuli that are perceived on some trials and not on
736 others, for example at masking or rivalry threshold, and fit decoders for one
737 shared stimulus quantity independently on two regions that both encode it,
738 selected by the decodability criterion of Section 4 on separate data. Clas-
739 sify trials by perceptual state with a no-report measure, such as reflexive
740 eye movements or pupil responses that track the dominant percept in rivalry
741 (Tsuchiya et al., 2015), so that the classification does not also select for the
742 neural correlates of reporting; report-based classification is a secondary anal-

743 ysis. Match the two classes on phase coherence between the regions and
744 on each region’s own decoding error (or, where the stimulus value is uncer-
745 tain, on each region’s signal strength). The statistic is the excess agreement
746 over a fidelity-matched null: pair region *A*’s decode on one trial with region
747 *B*’s decode on another trial of the same stimulus and error stratum, which
748 preserves both marginal accuracies and removes any coupling of the errors.

749 The proposed link predicts a larger excess on perceived trials at matched
750 coherence and matched single-region accuracy (Fig. 5). Equal excess across
751 classes, or excess that follows coherence and not perceptual state, counts
752 against it. A positive result is not yet specific: noise shared upstream of both
753 regions also correlates their errors, so the excess should be compared against
754 an upstream region’s decoding error, matched in the same way. A self-model
755 comparison has the same form: at matched accuracy for the scene variable,
756 the reconstruction of perceiver-relative variables, such as hand position rel-
757 ative to the target, from a declared local encoding should be more accurate
758 on perceived trials. These are correlational tests of proposed correlates, and
759 the no-report classification is itself an indirect marker of experience.

760 **10.3 What a test of the full proposal must establish**

761 A test needs separate estimates of effective coupling and diffusion, shared
762 content on overlapping regions, and reconstruction from a specified encoding
763 at a declared fidelity. The cover, relevant state family and decoder must
764 be fixed without selecting them for agreement. An independent experiential

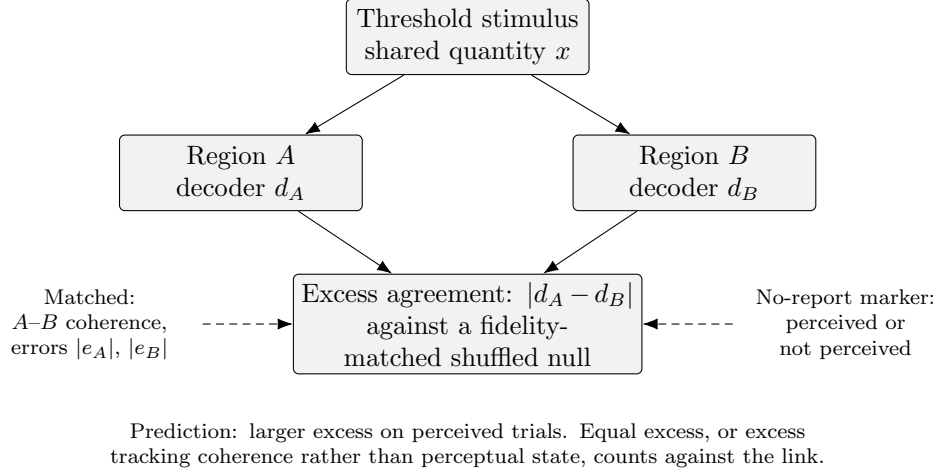


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

765 endpoint is needed to evaluate the interpretation; absence of a report alone
 766 does not define absence of experience. A unified episode with incompati-
 767 ble decoded descriptions would challenge the unity identification. A unified
 768 episode lacking the proposed self-encoding would challenge the minimal-self
 769 claim. Conversely, satisfying the mathematical conditions without such an
 770 episode would challenge their sufficiency as correlates.

771 For the physical route, cortical actuator modes and prices, the geometry–
 772 coupling relation and a dynamical onset model remain to be measured. The
 773 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
 774 diction. Controlled changes in coordination and independently decoded con-

775 tent agreement would address the representational connection. The synthetic
776 decoder exercise calibrates part of these measurements, and the EEG exercise
777 identifies what a pooled bipolar estimand cannot measure; neither identifies
778 a neural content code or tests experience.

779 10.4 Explanatory scope

780 Integrated-information, global-workspace and predictive-processing accounts
781 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
782 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
783 relations among coupling, local content and reconstruction. Its formal field
784 condition requires a continuous kernel, a normalized substrate, a specified
785 average coupling and noise, and a site-contribution bound. Those conditions
786 do not identify an electromagnetic substrate; the extracellular interpretation
787 needs the separate calibration of Section 9. The joint witness shows that
788 the formal hypotheses can hold together in a declared model. A cortical
789 realization and an experiential interpretation require additional evidence.

790 Four limitations bound every claim above. Synchrony is not itself an
791 experiential measure, and nothing here establishes a consciousness criterion:
792 the conditions are proposed correlates to be tested against report or another
793 independent endpoint. The present EEG analysis checks a measurement
794 pipeline and does not validate any experiential claim. The formal results
795 are conditional on declared encoders, covers and maps whose cortical coun-
796 terparts are unmeasured. The cover on which compatibility is assessed is

797 fixed only relative to pre-registered variables, decoder class and criterion,
798 and the phase-to-content bounds are stated for spatial covers rather than for
799 the content cover that criterion yields.

800 The remaining explanatory question is whether these relations occur in
801 neural activity and matter for experience. The framework identifies candidate
802 correlates and tests of their physical and representational premises. It does
803 not explain why satisfying them would be accompanied by experience.

804 11 Conclusions

805 Phase coherence, compatibility and self-reconstruction are separable condi-
806 tions. Coherence constrains shared content only through encoders shared on
807 overlaps and with a bound that grows as the square root of the population
808 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
809 order; chaining it through overlapping patches reaches 0.312 mm of that
810 sheet, and a spectral route removes the population count only for a declared
811 connectivity at a frequency-locked configuration. Across the centimetre sep-
812 arations of distinct cortical territories, the positive link from coherence to
813 content is the bridge assumption E78 rather than a derived result. Com-
814 patible local descriptions, not synchrony, determine one global state; and a
815 contracting self-map fixes that state while reconstructing only a family of
816 bounded diameter. Each result replaces an assumption about synchrony by
817 a measurement: independently decoded agreement on overlaps, a declared

818 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
819 ison of Section 10.2 states how such measurements could count against the
820 proposed link to conscious content.

821 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits

Table 2 (continued)

Symbol	Defined	Meaning
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	Section 9	Extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

822 Data and code availability

823 The Lean development, simulation scripts, tests and saved numerical sum-
824 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
825 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-

lib commit 5ce203ecd53f8c3e47ee5530d7c86b94a55f0183. Publication
simulation values are generated from saved summaries without rerunning
production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
and selection are described in Supplemental Material and implemented in
simulations/empirical_collapse.py. The repository contains the repro-
ducible analysis and does not redistribute the raw recordings.

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Use of AI tools

AI assistance to this work was substantial and is not separable by task.
Several coding assistants and chat interfaces were used in an interleaved way
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Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them

845 they contributed to the Lean development, the simulation and analysis code,
846 and the drafting and revision of both documents. Because the tools were
847 used in combination and in alternation, no section, proof, figure or result is
848 attributable to any one of them, and none is claimed to be. The author is
849 responsible for all content, including everything a tool produced. Machine
850 checking establishes the stated Lean results under their hypotheses; it does
851 not validate biological interpretations or replace scientific review of generated
852 code and prose.

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